

δ -rings and λ -rings

André Joyal

Presented by P. Ribenboim, F.R.S.C.

Translated by F. Murphy-Hernandez

Abstract

We show how to obtain the theory of λ -rings [3] from that of δ -rings [4]. The sequence of operations $(\lambda_n \mid n \in \mathbb{N})$ is replaced by a sequence of operations $(\delta_p \mid p \text{ prime})$ satisfying explicit identities. Consequently, we can exhibit for each $n \in \mathbb{N}$ a remarkable basis of the characters of the symmetric group S_n .

1 δ -operations on a λ -ring

Let $(\psi^n \mid n \geq 1)$ be the sequence of Adams operations in the theory of λ -rings. We know that ψ^n is a ring endomorphism and that

$$\psi^n \circ \psi^m = \psi^{nm} \quad \text{for all } n, m \geq 1.$$

For each prime number p , let c^p be the cyclic power operation of order p [1]. We verify the identity

$$p c^p(x) = x^p + (p-1)\psi^p(x).$$

Set

$$\delta_p(x) = \psi^p(x) - c^p(x).$$

Then we have

$$\psi^p(x) = x^p + p \delta_p(x).$$

Since the free λ -ring on one generator is p -torsion-free and ψ^p is an endomorphism, we conclude that the operation δ_p endows this ring with a δ -ring structure [4]. More generally, every λ -ring is a δ -ring with respect to each prime number p . We wish to make explicit the relations between δ_{p_1} and δ_{p_2} for distinct primes p_1 and p_2 .

Let $q_1 = p_1^{r_1}$ and $q_2 = p_2^{r_2}$, where p_1 and p_2 are distinct primes. Let A be a commutative ring equipped with two unary operations δ_1 and δ_2 . Suppose that (A, δ_1) is a δ -ring with respect to (p_1, q_1) and that (A, δ_2) is a δ -ring with respect to (p_2, q_2) .

Definition 1. *The operations δ_1 and δ_2 are permutable if the following identity is satisfied:*

$$\begin{aligned} \delta_1(\delta_2(x)) + \frac{q_2^{-1}}{p_1^{-1} - 1} \delta_1(x)^{q_2} + \frac{q_2^{-1}}{1 = 1} \frac{1}{p_2(1)} p_1^{-1-1} \delta_1(x)^1 q_1(q_2^{-1}) \\ = \delta_2(\delta_1(x)) + \frac{q_1^{-1}}{p_2^{-1} - 1} \delta_2(x)^{q_1} + \frac{q_1^{-1}}{1 = 1} \frac{1}{p_1(1)} p_2^{-1-1} \delta_2(x)^1 q_2(q_1^{-1}) \end{aligned}$$

The permutability of the operations δ_1 and δ_2 implies the commutation of the Frobenius endomorphisms

$$f_1(x) = x^{q_1} + p_1\delta_1(x), \quad f_2(x) = x^{q_2} + p_2\delta_2(x).$$

Conversely, if A is p_1p_2 -torsion-free, the relation $f_1f_2 = f_2f_1$ implies the permutability of δ_1 and δ_2 .

2 P -rings

Let P be a set of prime numbers.

Definition 2. A P -ring A is a commutative ring equipped with a unary operation $\delta_p : A \rightarrow A$ for each $p \in P$. The following conditions are required:

1. A equipped with δ_p is a δ -ring with respect to the pair (p, p) ;
2. δ_p and δ_ℓ are permutable for every pair of distinct elements $p, \ell \in P$.

Example 3. Every λ -ring is naturally equipped with a P -ring structure where P is the set of all prime numbers.

Example 4. Let $n \geq 1$ be an integer and let $\zeta_n \in \mathbb{C}$ be a primitive n -th root of unity. Let R_n be the ring of integers of the cyclotomic field $\mathbb{Q}(\zeta_n)$. Let P_n be the set of primes not dividing n . For each $p \in P_n$, let f_p be the unique automorphism of R_n such that $f_p(\zeta_n) = \zeta_n^p$. For all $x \in R_n$, we have

$$f_p(x) \equiv x^p \pmod{pR_n}.$$

This shows that R_n is a δ -ring with respect to the pair (p, p) . Since $f_p f_\ell = f_\ell f_p$ for all $p, \ell \in P_n$, we conclude that R_n is a P -ring.

A morphism of P -rings is a ring homomorphism which preserves each operation δ_p ($p \in P$). We verify that every Frobenius endomorphism f_p ($p \in P$) is a morphism of P -rings.

Let A be a P -ring and let $M(P)$ be the set of finite sequences of elements of P . For each $\sigma = (p_1, \dots, p_n) \in M(P)$, set

$$\delta_\sigma = \delta_{p_1} \circ \delta_{p_2} \circ \dots \circ \delta_{p_n}.$$

Denote by $C(P) \subset M(P)$ the subset of increasing sequences with respect to the natural order on P .

Theorem 5. Let $A = \mathbb{Z}[x_\sigma \mid \sigma \in C(P)]$ be the polynomial ring over indeterminates x_σ ($\sigma \in C(P)$). There is a unique P -ring structure on A such that $\delta_p(x_\sigma) = x_{p\sigma}$ for all $\sigma \in C(P)$. Equipped with this structure, A is the free P -ring on $x_\emptyset$.

Denote by $\mathcal{A}$ the category of rings and by $\mathcal{PA}$ the category of P -rings.

Theorem 6. The forgetful functor $U : \mathcal{PA} \rightarrow \mathcal{A}$ has a right adjoint $V : \mathcal{A} \rightarrow \mathcal{PA}$.

Corollary 7. The functor V is representable by the ring $\mathbb{Z}[x_\sigma \mid \sigma \in C(P)]$. There is a natural bijection

$$V(A) \cong A^{C(P)}.$$

3 λ -rings

In what follows, we assume that P is the set of all prime numbers. For every integer $n \geq 1$, we define a Frobenius endomorphism

$$f_n = f_{p_1}^{r_1} \cdots f_{p_k}^{r_k},$$

where $p_1 \cdots p_k$ is a decomposition of n into prime factors.

Proposition 8. *In the theory of P -rings, one can define a unique sequence of unary operations $\alpha_1, \alpha_2, \dots$ satisfying the identities*

$$f_n(x) = \sum_{r|n} \alpha_r(x)^{n/r} \quad (n \geq 1).$$

Proposition 9. *There is a unique P -ring structure on $\mathbb{Z}[A_1, A_2, \dots]$ such that $\alpha_n(A_1) = A_n$ for all $n \geq 1$. Equipped with this structure, $\mathbb{Z}[A_1, A_2, \dots]$ is a free P -ring on A_1 .*

Theorem 10. *When P is the set of all prime numbers, the concepts of P -rings and λ -rings are equivalent.*

The proof of this theorem uses the approach of Cartier [2].

Example 11. *Let R_n be the ring of Example 2. We can equip $R_n[1/n]$ with a λ -ring structure. Indeed, it suffices to define $f_p = \text{id}$ for $p \notin P_n$. In particular, $\mathbb{Z}[1/n]$ is a λ -ring.*

The free λ -ring $\Lambda[x]$ on one generator is a polynomial ring

$$\mathbb{Z}[x, \lambda^1 x, \lambda^2 x, \dots].$$

We introduce a natural grading on $\Lambda[x]$ by setting $\deg \lambda^n x = n$ for every $n \geq 1$. This graded ring can also be described using the character theory of the symmetric groups S_n ($n \geq 0$) [1]. Let $R(S_n)$ be the character group of S_n . Consider the inclusion $S_n \times S_m \subset S_{n+m}$. We define a bilinear operation

$$R(S_n) \times R(S_m) \rightarrow R(S_n \times S_m) \rightarrow R(S_{n+m})$$

and consequently, a graded ring structure on

$$\bigoplus_{n \geq 0} R(S_n).$$

This ring is isomorphic to $\Lambda[x]$ [5].

Now consider the polynomial ring $\mathbb{Z}[\delta_\sigma(x) \mid \sigma \in C(P)]$ mentioned in Theorem 1. We introduce a grading on this ring by setting

$$\deg \delta_\sigma(x) = p_1 p_2 \cdots p_n$$

whenever $\sigma = (p_1, p_2, \dots, p_n)$.

Corollary 12. *The graded rings $\bigoplus_{n \geq 0} R(S_n)$ and $\mathbb{Z}[\delta_\sigma(x) \mid \sigma \in C(P)]$ are isomorphic.*

References

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Department of Mathematics and Computer Science
University of Quebec at Montreal